

# Measures

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*Remark.* A measure assigns a size to each set in a [collection of sets](#): how many elements it has, how long it is, or how likely it is. This page builds measures from  $\sigma$ -algebras and additivity, using the [sets and functions](#) page's definitions. One kind of measure, the probability measure (Definition 0.19), is the starting point of the Morrison Lab's probability notes ([probability measure](#)).

## $\sigma$ -algebras

**Definition 0.1** ( $\sigma$ -algebra). A  $\sigma$ -algebra on a set  $S$  is a [collection](#)  $\mathcal{S}$  of [subsets](#) of  $S$  that satisfies:

- $\mathcal{S}$  contains  $S$  itself.
- For each set  $A$  in  $\mathcal{S}$ ,  $\mathcal{S}$  contains its [complement](#)  $S \setminus A$ .
- For each [sequence](#)  $A_1, A_2, \dots$  of sets in  $\mathcal{S}$ ,  $\mathcal{S}$  contains their union  $\bigcup_{i=1}^{\infty} A_i$ .

*Remark 0.1* ( $\sigma$ -field). Other sources call a  $\sigma$ -algebra a  $\sigma$ -**field** (see [Wikipedia:  \$\sigma\$ -algebra](#)). For example, Billingsley (1995) uses “ $\sigma$ -field”. The two names mean the same thing.

**Example 0.1** ( $\sigma$ -algebras for die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  ([sets of die rolls](#)), the collection of all subsets of  $D$  is a  $\sigma$ -algebra on  $D$ , since complements and unions of subsets of  $D$  are again subsets of  $D$ . So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$$

which contains  $D$ , the complement of each of its sets, and every union of its sets: for example,  $\{2, 4, 6\} \cup \{1, 3, 5\} = D$ .

**Example 0.2** (A collection that is not a  $\sigma$ -algebra). The collection  $\{\emptyset, \{1\}, D\}$  is not a  $\sigma$ -algebra on  $D = \{1, 2, 3, 4, 5, 6\}$ : it contains  $\{1\}$  but not its complement  $D \setminus \{1\} = \{2, 3, 4, 5, 6\}$ .

**Definition 0.2** (Closed under an operation (closure)). A set  $A$  is **closed under** an operation, such as addition of numbers or union of sets, if applying the operation to elements of  $A$  always gives an element of  $A$ . This property is called **closure** under the operation.

**Example 0.3** (Closed and not closed).

- The **even** integers are closed under addition: for example,  $2 + 4 = 6$  is even, and  $2m + 2n = 2(m + n)$  is even for all integers  $m$  and  $n$ .
- The odd integers are not closed under addition: 1 and 3 are odd, but  $1 + 3 = 4$  is even.
- A  **$\sigma$ -algebra**  $\mathcal{S}$  on a set  $S$  is closed under complements in  $S$  and under unions of sequences of its sets, because Definition 0.1 requires both. The collection in Example 0.2 is not closed under complements.

**Theorem 0.1** (Closure properties of a  $\sigma$ -algebra). If  $\mathcal{S}$  is a  **$\sigma$ -algebra** on a set  $S$ , then  $\mathcal{S}$  contains:

- the **empty set**  $\emptyset$ ;
- the union  $A_1 \cup \dots \cup A_n$  of any **finitely many** sets  $A_1, \dots, A_n$  in  $\mathcal{S}$ ;
- the intersection  $\bigcap_{i=1}^{\infty} A_i$  of any sequence  $A_1, A_2, \dots$  of sets in  $\mathcal{S}$ .

So  $\mathcal{S}$  is **closed under** finite unions and under intersections of sequences of its sets.

## **i** Proof

*Proof.* **Empty set.**  $\mathcal{S}$  contains  $S$ , so it contains the complement of  $S$ , which is  $S \setminus S = \emptyset$ . **Finite unions.** Extend  $A_1, \dots, A_n$  to a sequence by setting  $A_{n+1} = A_{n+2} = \dots = \emptyset$ ; each of these sets is in  $\mathcal{S}$ , by the first part. Then:

$$\begin{aligned} A_1 \cup \dots \cup A_n &= A_1 \cup \dots \cup A_n \cup \emptyset \cup \emptyset \cup \dots \quad (\text{a union with } \emptyset \text{ adds no elements}) \\ &= \bigcup_{i=1}^{\infty} A_i \quad (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and  $\mathcal{S}$  contains  $\bigcup_{i=1}^{\infty} A_i$  by the union rule of Definition 0.1.

**Countable intersections.** Each  $A_i$  is a subset of  $S$ , so an element of  $S$  is outside  $S \setminus A_i$

exactly when it is in  $A_i$ ; that is,  $S \setminus (S \setminus A_i) = A_i$ . So, by [De Morgan's laws](#):

$$\begin{aligned} S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) &= \bigcap_{i=1}^{\infty} (S \setminus (S \setminus A_i)) \quad (\text{De Morgan's laws}) \\ &= \bigcap_{i=1}^{\infty} A_i \quad (S \setminus (S \setminus A_i) = A_i) \end{aligned}$$

Each  $S \setminus A_i$  is in  $\mathcal{S}$  by the complement rule, so their union is in  $\mathcal{S}$  by the union rule, and the complement of that union is in  $\mathcal{S}$  by the complement rule again.  $\square$

**Theorem 0.2** (All subsets form a  $\sigma$ -algebra). *For any set  $S$ , the collection of all subsets of  $S$  is a  $\sigma$ -algebra on  $S$ .*

### Proof

*Proof.* Each condition of Definition [0.1](#) holds:

- $S$  is a subset of itself.
- For each subset  $A$  of  $S$ , the complement  $S \setminus A$  contains only elements of  $S$ , so it is a subset of  $S$ .
- For each sequence  $A_1, A_2, \dots$  of subsets of  $S$ , every element of  $\bigcup_{i=1}^{\infty} A_i$  is in some  $A_i$  and so in  $S$ ; the union is a subset of  $S$ .

$\square$

**Theorem 0.3** (An intersection of  $\sigma$ -algebras is a  $\sigma$ -algebra). *Let  $\mathcal{S}_j$  be a  $\sigma$ -algebra on a set  $S$  for each  $j$  in a nonempty set of indices  $J$ . Then their [intersection](#)  $\bigcap_{j \in J} \mathcal{S}_j$ , the collection of subsets of  $S$  that are in every  $\mathcal{S}_j$ , is a  $\sigma$ -algebra on  $S$ .*

### Proof

*Proof.* Each condition of Definition [0.1](#) holds:

- $S$  is in every  $\mathcal{S}_j$ , so it is in the intersection.
- If  $A$  is in the intersection, then  $A$  is in every  $\mathcal{S}_j$ , so its complement  $S \setminus A$  is in every  $\mathcal{S}_j$ , and so it is in the intersection.
- If  $A_1, A_2, \dots$  are in the intersection, then they are in every  $\mathcal{S}_j$ , so their union  $\bigcup_{i=1}^{\infty} A_i$  is in every  $\mathcal{S}_j$ , and so it is in the intersection.

$\square$

**Definition 0.3** ( $\sigma$ -algebra generated by a collection). Let  $\mathcal{C}$  be a **collection** of subsets of a set  $S$ . The  **$\sigma$ -algebra generated by  $\mathcal{C}$** , written  $\sigma(\mathcal{C})$ , is the intersection of all the  $\sigma$ -algebras on  $S$  that contain every set in  $\mathcal{C}$ .

*Remark 0.2* (The smallest  $\sigma$ -algebra containing  $\mathcal{C}$ ).  $\sigma(\mathcal{C})$  is a  $\sigma$ -algebra by Theorem 0.3 (the collection of all subsets of  $S$  is one of the  $\sigma$ -algebras intersected, by Theorem 0.2, so there is at least one). It contains every set in  $\mathcal{C}$ , and it is contained in every other  $\sigma$ -algebra on  $S$  that does, so it is the smallest  $\sigma$ -algebra on  $S$  that contains  $\mathcal{C}$ .

**Example 0.4** (The  $\sigma$ -algebra generated by the even rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$ , let  $\mathcal{C} = \{\{2, 4, 6\}\}$ . Every  $\sigma$ -algebra on  $D$  that contains  $\{2, 4, 6\}$  must also contain  $D$ , the complement  $\{1, 3, 5\}$ , and  $\emptyset$  (Definition 0.1 and Theorem 0.1). The collection  $\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$  is already a  $\sigma$ -algebra (Example 0.1), so it is  $\sigma(\mathcal{C})$ .

**Definition 0.4** (Borel  $\sigma$ -algebra). The **Borel  $\sigma$ -algebra** on  $\mathbb{R}$ , written  $\mathcal{B}$ , is the  $\sigma$ -algebra generated by (Definition 0.3) the collection of all **intervals**. Its sets are called **Borel sets**.

**Example 0.5** (Some Borel sets).

- Every interval, such as  $[0, 1]$ ,  $(2, 5)$ , or  $[0, \infty)$ , is a Borel set.
- A single point  $\{a\} = [a, a]$  is an interval, so it is a Borel set.
- The **integers** are a Borel set: they are the union of the sequence of single points  $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \dots$  (**countable and uncountable sets** lists them this way), and  $\mathcal{B}$  contains that union by Definition 0.1.
- The complement  $\mathbb{R} \setminus [0, 1]$  is a Borel set, because  $\mathcal{B}$  contains the complement of each of its sets.

## Pairwise disjoint sets

**Definition 0.5** (Pairwise disjoint sets (disjoint, mutually disjoint)). Finitely or countably many sets  $A_1, A_2, \dots$  are **pairwise disjoint** (also called **disjoint** or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

**Example 0.6** (Low and high die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), the sets  $\{1, 2\}$  and  $\{5, 6\}$  are pairwise disjoint: no roll is in both. The sets  $\{2, 4, 6\}$  and  $\{1, 2\}$  are not, because 2 is in both.

## Infinite sums

**Definition 0.6** (Non-decreasing sequence). A sequence  $a_1, a_2, \dots$  of real numbers or of extended non-negative reals is **non-decreasing** if each term is at least as large as the one before it:

$$a_n \leq a_{n+1} \quad \text{for every } n \in \mathbb{N}$$

**Example 0.7** (Non-decreasing and not).

- $1, 2, 2, 3, 3, 3, \dots$  is non-decreasing: a term may equal the one before it.
- $1, 2, \infty, \infty, \dots$  is non-decreasing, because  $\infty$  is greater than every real number.
- $1, \frac{1}{2}, \frac{1}{3}, \dots$  is not non-decreasing, because  $a_2 = \frac{1}{2} < 1 = a_1$ .

**Definition 0.7** (Partial sum). Let  $a_1, a_2, \dots$  be a sequence of real numbers or of extended non-negative reals. For each  $n \in \mathbb{N}$ , its  $n$ th **partial sum** is the sum of its first  $n$  terms:

$$s_n \stackrel{\text{def}}{=} \sum_{i=1}^n a_i = a_1 + \dots + a_n$$

**Example 0.8** (Partial sums of halves). For  $a_i = 1/2^i$ , the sequence  $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ , the first three partial sums are  $s_1 = \frac{1}{2}$ ,  $s_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ , and  $s_3 = \frac{3}{4} + \frac{1}{8} = \frac{7}{8}$ . In general,  $s_n = 1 - 1/2^n$ .

**Definition 0.8** (Infinite sum (series)). Let  $a_1, a_2, \dots$  be a sequence in  $[0, \infty]$ , with partial sums  $s_n$  (Definition 0.7). The **infinite sum** (also called a **series**)  $\sum_{i=1}^{\infty} a_i$  is the limit of the partial sums:

$$\sum_{i=1}^{\infty} a_i \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} s_n$$

Here the limit is a real number when the partial sums are real numbers that converge to it, and it is  $\infty$  when the partial sums diverge to  $\infty$  or some partial sum is  $\infty$ ; Lemma 0.1, next, shows that one of these always happens. Sums with other index ranges, such as

$\sum_{k=0}^{\infty} a_k$ , are defined the same way.

**Example 0.9** (A finite and an infinite sum).

- For  $a_i = 1/2^i$ , the partial sums  $s_n = 1 - 1/2^n$  (Example 0.8) converge to 1, because  $1/2^n \rightarrow 0$ . So  $\sum_{i=1}^{\infty} 1/2^i = 1$ .
- For  $a_i = 1$ , the partial sums are  $s_n = n$ , which diverge to  $\infty$ . So  $\sum_{i=1}^{\infty} 1 = \infty$ .

**Lemma 0.1** (Sums of non-negative terms). *Let  $a_1, a_2, \dots$  be values in  $[0, \infty]$ , with partial sums  $s_n$  (Definition 0.7), where  $x + \infty = \infty$  for every  $x$  in  $[0, \infty]$ . Then the partial sums are **non-decreasing**,  $s_1 \leq s_2 \leq \dots$ , so their **limit**, the infinite sum  $\sum_{i=1}^{\infty} a_i$  (Definition 0.8), always exists, and it is either a finite number or  $\infty$ .*

### Proof

*Proof.* For each  $n$ :

$$\begin{aligned} s_{n+1} &= s_n + a_{n+1} && \text{(definition of } s_{n+1}) \\ &\geq s_n && (a_{n+1} \geq 0) \end{aligned}$$

If some partial sum  $s_N$  is  $\infty$ , then  $s_n = \infty$  for every  $n \geq N$ , so the limit is  $\infty$ . Otherwise, the partial sums form a non-decreasing sequence of **real numbers**. If that sequence is **bounded above**, it **converges** to a finite number (see [Wikipedia: Monotone convergence theorem](#)). If it is not bounded above, then for every number  $M$  some  $s_N$  exceeds  $M$ , and so does every later  $s_n \geq s_N$ ; the limit is  $\infty$ .  $\square$

## Additivity

**Definition 0.9** (Finite additivity). Let  $\mathcal{S}$  be a  **$\sigma$ -algebra** on a set  $S$ . A **function**  $\mu : \mathcal{S} \rightarrow [0, \infty]$ , with values in the **extended non-negative reals**, is **finitely additive** if, for every finite collection of **pairwise disjoint** sets  $A_1, \dots, A_n$  in  $\mathcal{S}$ , the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

**Example 0.10** (Counting elements is finitely additive). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), let  $\mu(A) \stackrel{\text{def}}{=} |A|$ , the number of elements of  $A$  (cardinality), for each set  $A$  in the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1). The sets  $\{1, 2\}$  and  $\{5, 6\}$  are pairwise disjoint (Example 0.6), and:

$$\begin{aligned}\mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\ &= 4 && \text{(count the elements)} \\ &= 2 + 2 && \text{(write 4 as a sum)} \\ &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each set's elements)}\end{aligned}$$

The same holds for any pairwise disjoint sets, because the sizes of disjoint sets add, so  $\mu$  is finitely additive.

**Example 0.11** (Squaring the count is not finitely additive). On the same  $\sigma$ -algebra, let  $\nu(A) \stackrel{\text{def}}{=} |A|^2$ . The sets  $\{1\}$  and  $\{2\}$  are pairwise disjoint, but

$$\begin{aligned}\nu(\{1\} \cup \{2\}) &= \nu(\{1, 2\}) && \text{(take the union)} \\ &= 2^2 \\ &= 4, && \text{(count, then square)} \\ \nu(\{1\}) + \nu(\{2\}) &= 1^2 + 1^2 \\ &= 2, && \text{(count each set, then square)}\end{aligned}$$

and  $4 \neq 2$ , so  $\nu$  is not finitely additive.

**Definition 0.10** (Countable additivity ( $\sigma$ -additivity)). Let  $\mathcal{S}$  be a  $\sigma$ -algebra on a set  $S$ . A function  $\mu : \mathcal{S} \rightarrow [0, \infty]$  is **countably additive** (also called  **$\sigma$ -additive**) if, for every sequence of pairwise disjoint sets  $A_1, A_2, \dots$  in  $\mathcal{S}$ , the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

**Example 0.12** (Counting elements is countably additive). For the counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.10, take any sequence of pairwise disjoint sets  $A_1, A_2, \dots$ . No two of them share an element, and  $D$  has only six elements, so at most six of the  $A_i$  contain any elements; the rest are  $\emptyset$ , with  $\mu(\emptyset) = 0$ . The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.10) shows that those terms add up to  $\mu$  of the union. So  $\mu$  is countably additive.

*Remark 0.3* (The sum in countable additivity always has a value). By Lemma 0.1, the right-hand side of the equation in Definition 0.10,  $\sum_{i=1}^{\infty} \mu(A_i)$ , always has a value, because each  $\mu(A_i)$  is in  $[0, \infty]$ . For example, if  $\mu(A_i) = 1/2^i$  for each  $i$ , the partial sums are  $s_n = 1 - 1/2^n$ , and the sum is 1; if  $\mu(A_i) = 1$  for each  $i$ , the partial sums are  $s_n = n$ , and the sum is  $\infty$ .

**Lemma 0.2** (Countable additivity and the empty set). *If  $\mu$  is a countably additive function on a  $\sigma$ -algebra  $\mathcal{S}$ , then  $\mu(\emptyset) = 0$  or  $\mu(\emptyset) = \infty$ .*

**i** Proof

*Proof.* The set  $\emptyset = S \setminus S$  is in  $\mathcal{S}$ , as the complement of  $S$ , and the sequence  $\emptyset, \emptyset, \dots$  is pairwise disjoint with union  $\emptyset$ , so:

$$\begin{aligned} \mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) \quad (\text{the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) \quad (\text{countable additivity}) \end{aligned}$$

If  $\mu(\emptyset) = c$  for a finite  $c > 0$ , the right-hand side is  $c + c + \dots = \infty \neq c$ , a contradiction. So  $\mu(\emptyset)$  is 0 or  $\infty$ .  $\square$

**Example 0.13** (A countably additive function with  $\mu(\emptyset) = 0$ ). The counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.12 is countably additive, and:

$$\begin{aligned} \mu(\emptyset) &= |\emptyset| \quad (\text{definition of } \mu) \\ &= 0 \quad (\emptyset \text{ has no elements}) \end{aligned}$$

**Example 0.14** (A countably additive function with  $\mu(\emptyset) = \infty$ ). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$  (sets of die rolls), let  $\mu(A) \stackrel{\text{def}}{=} \infty$  for every set  $A$  in the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1), including  $A = \emptyset$ . For any sequence of pairwise disjoint sets  $A_1, A_2, \dots$ , the left-hand side of the countable additivity equation is:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \infty \quad (\text{definition of } \mu)$$

and the right-hand side, the limit of its partial sums (Lemma 0.1), is:

$$\begin{aligned}\sum_{i=1}^{\infty} \mu(A_i) &= \infty + \infty + \dots \quad (\text{definition of } \mu) \\ &= \infty \quad (\text{every partial sum is } \infty)\end{aligned}$$

The two sides agree, so  $\mu$  is countably additive, and  $\mu(\emptyset) = \infty$ .

*Remark 0.4* ( $\mu(\emptyset) = 0$  is an extra requirement). Example 0.13 and Example 0.14 show that both values allowed by Lemma 0.2 occur for countably additive functions. So  $\mu(\emptyset) = 0$  is an extra requirement, not a consequence of countable additivity.

**Theorem 0.4** (Countable additivity implies finite additivity). *If  $\mu$  is a countably additive function on a  $\sigma$ -algebra  $\mathcal{S}$ , and  $\mu(\emptyset) = 0$ , then  $\mu$  is finitely additive.*

### **i** Proof

*Proof.* Let  $A_1, \dots, A_n$  be pairwise disjoint sets in  $\mathcal{S}$ , and extend them to a sequence by setting  $A_{n+1} = A_{n+2} = \dots = \emptyset$ . The extended sequence is still pairwise disjoint, since  $\emptyset$  shares no element with any set, and its union is  $A_1 \cup \dots \cup A_n$ . So:

$$\begin{aligned}\mu(A_1 \cup \dots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^{\infty} \mu(A_i) && (\text{countable additivity}) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^n \mu(A_i) && (\mu(\emptyset) = 0)\end{aligned}$$

□

**Example 0.15** (Finitely additive but not countably additive). Let  $S = \{0, 1, 2, \dots\}$ , with the  $\sigma$ -algebra of all subsets of  $S$  (Theorem 0.2), and define:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

$\mu$  is finitely additive. Let  $A_1, \dots, A_n$  be pairwise disjoint subsets of  $S$ . If every  $A_i$  is finite, then so is their union, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= 0 && \text{(a finite union of finite sets is finite)} \\
&= \sum_{i=1}^n 0 && \text{(a sum of zeros is 0)} \\
&= \sum_{i=1}^n \mu(A_i) && \text{(each } A_i \text{ is finite)}
\end{aligned}$$

If some  $A_k$  is infinite, then the union, which contains  $A_k$ , is infinite too, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= \infty && \text{(the union is infinite)} \\
&= \mu(A_k) + \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \mu(A_i) && (\mu(A_k) = \infty, \text{ and } \infty + x = \infty) \\
&= \sum_{i=1}^n \mu(A_i) && \text{(regroup the terms)}
\end{aligned}$$

$\mu$  is not countably additive. The single-element sets  $\{0\}, \{1\}, \{2\}, \dots$  are pairwise disjoint, and their union is  $S$ , which is infinite. So:

$$\begin{aligned}
\mu\left(\bigcup_{k=0}^{\infty} \{k\}\right) &= \mu(S) && \text{(the union is } S) \\
&= \infty && (S \text{ is infinite})
\end{aligned}$$

but:

$$\begin{aligned}
\sum_{k=0}^{\infty} \mu(\{k\}) &= 0 + 0 + \dots && \text{(each } \{k\} \text{ is finite)} \\
&= 0 && \text{(every partial sum is 0)}
\end{aligned}$$

*Remark 0.5* (Finite additivity does not imply countable additivity). Example 0.15 shows that the converse of Theorem 0.4 fails, even when  $\mu(\emptyset) = 0$ : a finitely additive function need not be countably additive (see [Wikipedia: Sigma-additive set function](#), “An additive function which is not  $\sigma$ -additive”). The function  $\mu$  in Example 0.15 has  $\mu(\emptyset) = 0$ , because  $\emptyset$  is finite.

## Measures

**Definition 0.11** (Measure). A **measure** on a set  $S$  with a  $\sigma$ -algebra  $\mathcal{S}$  is a function  $\mu : \mathcal{S} \rightarrow [0, \infty]$  that satisfies:

- $\mu(\emptyset) = 0$ .
- $\mu$  is **countably additive**.

**Example 0.16** (Counting elements is a measure). The counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.10, defined on the  $\sigma$ -algebra of all subsets of  $D$  (Example 0.1), takes values in  $[0, \infty]$ , is countably additive (Example 0.12), and gives  $\mu(\emptyset) = 0$ , so it is a measure.

**Corollary 0.1** (Measures are finitely additive). *Every measure is finitely additive.*

### Proof

*Proof.* A measure  $\mu$  is countably additive and has  $\mu(\emptyset) = 0$  (Definition 0.11), so Theorem 0.4 applies to it.  $\square$

**Definition 0.12** (Counting measure). The **counting measure** on a set  $S$  is the **measure** on the  $\sigma$ -algebra of all subsets of  $S$  that assigns each finite set its number of elements, and each infinite set the value  $\infty$ :

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

**Example 0.17** (Counting measure on the non-negative integers). For the counting measure  $\mu$  on the **non-negative integers**  $\{0, 1, 2, \dots\}$ ,  $\mu(\{0, 1, 2\}) = 3$ , and the set of even numbers has  $\mu(\{0, 2, 4, \dots\}) = \infty$ . The even numbers are the union of the pairwise disjoint sets  $\{0\}, \{2\}, \{4\}, \dots$ , and countable additivity agrees:  $\sum_{k=0}^{\infty} \mu(\{2k\}) = 1 + 1 + \dots = \infty$ .

**Definition 0.13** (Measurable space). A **measurable space**  $(S, \mathcal{S})$  is a set  $S$  together with a  $\sigma$ -algebra  $\mathcal{S}$  on  $S$ ; the sets in  $\mathcal{S}$  are its **measurable sets**.

**Example 0.18** (Measurable spaces of die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$ :

- $(D, \mathcal{S})$ , with  $\mathcal{S}$  the collection of all subsets of  $D$ , is a measurable space (Example 0.1).
- $(D, \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\})$  is also a measurable space. Its measurable sets are only those four; for example,  $\{1\}$  is not one of them.

**Definition 0.14** (Measure space). A **measure space**  $(S, \mathcal{S}, \mu)$  is a **measurable space** together with a **measure**  $\mu$  on  $\mathcal{S}$ .

**Example 0.19** (Measure space of die rolls). For the die rolls  $D = \{1, 2, 3, 4, 5, 6\}$ , adding the counting function  $\mu(A) = |A|$  of Example 0.16 to the **measurable space**  $(D, \mathcal{S})$  of Example 0.18 makes  $(D, \mathcal{S}, \mu)$  a measure space.

**Definition 0.15** (Lebesgue measure). **Lebesgue measure** on  $\mathbb{R}$  is the **measure**  $\lambda$  on the **Borel  $\sigma$ -algebra**  $\mathcal{B}$  that assigns each **interval** its length: for real numbers  $a \leq b$ ,

$$\begin{aligned}\lambda([a, b]) &= \lambda((a, b)) \\ &= \lambda([a, b)) \\ &= \lambda((a, b]) \\ &\stackrel{\text{def}}{=} b - a,\end{aligned}$$

and  $\lambda(I) \stackrel{\text{def}}{=} \infty$  for each unbounded interval  $I$ .

**Lebesgue measure on an interval**  $I$ , such as  $[0, 1]$ , is  $\lambda$  applied only to the Borel sets that are subsets of  $I$ .

*Remark 0.6* (Lebesgue measure exists). Exactly one measure on  $\mathcal{B}$  assigns every interval its length; constructing it takes more measure theory than these notes cover (see [Wikipedia: Lebesgue measure](#), which also extends  $\lambda$  to a larger  $\sigma$ -algebra, the Lebesgue-measurable sets, with the same values on Borel sets).

**Example 0.20** (Lengths of some Borel sets).

- $\lambda([2, 5]) = 5 - 2 = 3$ .
- A single point has length zero:  $\lambda(\{a\}) = \lambda([a, a]) = a - a = 0$ .
- The integers are the union of the pairwise disjoint single points  $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \dots$  (Example 0.5), so countable additivity gives  $\lambda(\mathbb{Z}) = 0 + 0 + \dots = 0$ .
- $\lambda([0, \infty)) = \infty$ , because  $[0, \infty)$  is unbounded.
- Lebesgue measure on  $[0, 1]$  gives the whole interval  $[0, 1]$  the value  $\lambda([0, 1]) = 1 - 0 = 1$ .

*Remark 0.7* (Measures generalize size). A **measure** generalizes size: it can measure how many elements a set has, as in Definition 0.12, or how long a set of real numbers is, as Lebesgue measure (Definition 0.15) does, assigning each **interval**  $[a, b]$  its length  $b - a$ .

For example, the counting measure on  $\{0, 1, 2, \dots\}$  gives  $\{2, 3, 4, 5\}$  the value 4, while the Lebesgue measure of the interval  $[2, 5]$  is  $5 - 2 = 3$ . Both can serve as the measures  $\mu_1$  and  $\mu_2$  in the [Fubini–Tonelli theorem](#).

**Definition 0.16** (Measure zero (null set)). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#). A set  $A \in \mathcal{S}$  has **measure zero** (also called a **null set** for  $\mu$ ) if  $\mu(A) = 0$ .

**Example 0.21** (Sets of measure zero).

- For a [counting measure](#), only  $\emptyset$  has measure zero: a nonempty set has at least one element, so its counting measure is at least 1.
- For Lebesgue measure, every single point  $\{a\}$  and the integers  $\mathbb{Z}$  have measure zero (Example 0.20), although neither set is empty.

**Definition 0.17** (Almost everywhere ( $\mu$ -almost everywhere,  $\mu$ -a.e.)). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#). A statement about the elements of  $S$  holds  **$\mu$ -almost everywhere** (written  $\mu$ -a.e., or **almost everywhere** when  $\mu$  is clear from context) if the set of elements for which it fails is a [subset](#) of a set of [measure zero](#).

**Example 0.22** (Non-negative except on the integers). Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  with  $g(x) = -1$  when  $x$  is an integer and  $g(x) = x^2$  otherwise. The statement “ $g(x) \geq 0$ ” fails exactly for  $x \in \mathbb{Z}$ , and  $\lambda(\mathbb{Z}) = 0$  (Example 0.20), so  $g \geq 0$  holds  $\lambda$ -almost everywhere, though not everywhere.

For a counting measure, only  $\emptyset$  has measure zero (Example 0.21), so a statement holds almost everywhere only if it holds everywhere.

**Definition 0.18** (Finite and  $\sigma$ -finite measures). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#). The measure  $\mu$  is **finite** if  $\mu(S) < \infty$ . It is  **$\sigma$ -finite** if  $S$  is the union  $\bigcup_{i=1}^{\infty} A_i$  of a sequence of sets  $A_1, A_2, \dots$  in  $\mathcal{S}$  with  $\mu(A_i) < \infty$  for every  $i$ .

**Example 0.23** (Finite and  $\sigma$ -finite measures).

- Every finite measure is  $\sigma$ -finite: take  $A_1 = S$  and  $A_2 = A_3 = \dots = \emptyset$ , with  $\mu(A_1) = \mu(S) < \infty$  and  $\mu(\emptyset) = 0$ .
- The counting measure on the die rolls  $D$  is finite, with  $\mu(D) = 6$ .
- The counting measure on  $\{0, 1, 2, \dots\}$  is not finite, but it is  $\sigma$ -finite: it is the union of the sets  $A_i = \{0, 1, \dots, i\}$ , each with  $\mu(A_i) = i + 1 < \infty$ .
- Lebesgue measure on  $\mathbb{R}$  is not finite, since  $\lambda(\mathbb{R}) = \infty$ , but it is  $\sigma$ -finite:  $\mathbb{R}$  is the

union of the intervals  $A_i = [-i, i]$ , each with  $\lambda(A_i) = 2i < \infty$ .

## Probability measures

**Definition 0.19** (Probability measure). A **probability measure** on a measurable space  $(\Omega, \mathcal{F})$  is a **measure**  $P$  on  $\mathcal{F}$  that gives the whole set  $\Omega$  the value 1:

$$P(\Omega) = 1$$

Then  $(\Omega, \mathcal{F}, P)$  is a **probability space**,  $\Omega$  is its **sample space**, the elements of  $\Omega$  are **outcomes**, and the sets in  $\mathcal{F}$  are **events**.

**Example 0.24** (A fair die, and the unit interval).

- For one roll of a fair die, the sample space is  $\Omega = D = \{1, 2, 3, 4, 5, 6\}$ , the events are all subsets of  $D$ , and  $P(A) \stackrel{\text{def}}{=} |A|/6$ .  $P$  is the counting function  $\mu(A) = |A|$  of Example 0.16 divided by 6. So  $P(\emptyset) = 0$ , and  $P$  is countably additive, because each sum in Example 0.12 has at most six nonzero terms, and dividing a finite sum by 6 divides each term by 6. It is a probability measure, since  $P(D) = 6/6 = 1$ . The outcome 3 is in the event “the roll is odd”,  $\{1, 3, 5\}$ , which has  $P(\{1, 3, 5\}) = 3/6 = 1/2$ .
- Lebesgue measure on  $[0, 1]$  is a probability measure, because  $\lambda([0, 1]) = 1$  (Example 0.20).
- The counting measure on  $D$  is not a probability measure, because it gives  $D$  the value 6.

*Remark 0.8* (Mutually exclusive events). In probability, **pairwise disjoint events** are usually called **mutually exclusive**. For example, for one roll of a six-sided die, the events “the roll is **even**”,  $\{2, 4, 6\}$ , and “the roll is odd”,  $\{1, 3, 5\}$ , are mutually exclusive: no roll is both even and odd.

## Measurable functions and integrals

**Definition 0.20** (Measurable function). Let  $(S, \mathcal{S})$  be a measurable space. A function  $f : S \rightarrow \mathbb{R}$  is **measurable** (with respect to  $\mathcal{S}$ ) if, for every real number  $c$ , the set of elements where  $f$  is at most  $c$  is measurable:

$$\{s \in S : f(s) \leq c\} \in \mathcal{S}$$

*Remark 0.9* (Measurable functions in general). Other sources define a function  $f : S \rightarrow T$  between measurable spaces  $(S, \mathcal{S})$  and  $(T, \mathcal{T})$  to be measurable when  $\{s \in S : f(s) \in B\} \in \mathcal{S}$  for every  $B \in \mathcal{T}$ . For  $T = \mathbb{R}$  with the [Borel  \$\sigma\$ -algebra](#), that definition is equivalent to Definition 0.20 (see [Wikipedia: Measurable function](#)). For example, with  $B = (-\infty, c]$ , the general definition requires  $\{s \in S : f(s) \leq c\} \in \mathcal{S}$ , which is the condition in Definition 0.20.

**Example 0.25** (Measurable and not measurable). On the die rolls  $D$ , take the  $\sigma$ -algebra  $\mathcal{S} = \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$  of Example 0.1.

- Let  $f(x) = 1$  for even  $x$  and  $f(x) = 0$  for odd  $x$ . The set  $\{x \in D : f(x) \leq c\}$  is  $\emptyset$  when  $c < 0$ ,  $\{1, 3, 5\}$  when  $0 \leq c < 1$ , and  $D$  when  $c \geq 1$ . All three are in  $\mathcal{S}$ , so  $f$  is measurable.
- Let  $g(x) = x$ . Then  $\{x \in D : g(x) \leq 1\} = \{1\}$ , which is not in  $\mathcal{S}$ , so  $g$  is not measurable.

With the  $\sigma$ -algebra of all subsets of  $D$  instead, every set  $\{x \in D : g(x) \leq c\}$  is a subset of  $D$ , so every function  $D \rightarrow \mathbb{R}$ , including  $g$ , is measurable.

**Definition 0.21** (Integral of a non-negative function). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#), and let  $f : S \rightarrow [0, \infty)$  be [measurable](#). The **integral** of  $f$  with respect to  $\mu$  is

$$\int_S f d\mu \stackrel{\text{def}}{=} \sup \left\{ \sum_{i=1}^n \left( \inf_{s \in A_i} f(s) \right) \mu(A_i) \right\}$$

where the [supremum](#) is over all ways of writing  $S$  as the union of finitely many nonempty [pairwise disjoint](#) sets  $A_1, \dots, A_n$  in  $\mathcal{S}$ , and each  $\inf_{s \in A_i} f(s)$  is the [infimum](#) of the values of  $f$  on  $A_i$ . In the sums,  $0 \cdot \infty \stackrel{\text{def}}{=} 0$ , and  $x \cdot \infty \stackrel{\text{def}}{=} \infty$  for  $x > 0$ ; the supremum is  $\infty$  if some sum is  $\infty$ . The integral is also written  $\int_S f(s) d\mu(s)$ .

(Billingsley 1995, sec. 15)

*Remark 0.10* (Each sum is a lower estimate). Each sum in Definition 0.21 splits  $S$  into pieces and weights each piece  $A_i$  by the smallest value  $f$  approaches on it, so each sum is at most the integral. For Lebesgue measure on a closed interval  $[a, b]$  and a function  $f \geq 0$  that has a [Riemann integral](#) on  $[a, b]$ , the two integrals agree:  $\int_{[a,b]} f d\lambda = \int_a^b f(x) dx$  (see [Wikipedia: Lebesgue integral](#)).

**Example 0.26** (Integrals with respect to counting measure). Let  $\mu$  be the counting measure on the die rolls  $D$ , and let  $g : D \rightarrow [0, \infty)$ . Every function on  $D$  is measurable (Example 0.25).

*The integral is the sum of the values.* For any split of  $D$  into nonempty pairwise disjoint sets  $A_1, \dots, A_n$ :

$$\begin{aligned} \sum_{i=1}^n \left( \inf_{x \in A_i} g(x) \right) \mu(A_i) &= \sum_{i=1}^n \sum_{x \in A_i} \inf_{y \in A_i} g(y) \quad (\mu(A_i) \text{ counts the elements of } A_i) \\ &\leq \sum_{i=1}^n \sum_{x \in A_i} g(x) \quad (\text{an infimum is at most each value}) \\ &= \sum_{x \in D} g(x) \quad (\text{each } x \in D \text{ is in exactly one } A_i) \end{aligned}$$

The split into the six single points  $\{1\}, \dots, \{6\}$  gives exactly  $\sum_{x \in D} g(x)$ , so the supremum is  $\int_D g d\mu = \sum_{x \in D} g(x)$ . For example, with  $g(x) = x$ ,  $\int_D g d\mu = 1 + 2 + \dots + 6 = 21$ .

*An infinite integral.* For the counting measure  $\mu$  on  $\mathbb{N}$  and the constant function  $h(n) = 1$ , the split with one set,  $A_1 = \mathbb{N}$ , gives the sum  $1 \cdot \mu(\mathbb{N}) = 1 \cdot \infty = \infty$ , so  $\int_{\mathbb{N}} h d\mu = \infty$ .

**Definition 0.22** (Integral of a real-valued function). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#), and let  $f : S \rightarrow \mathbb{R}$  be [measurable](#). The **positive part** and **negative part** of  $f$  are the non-negative functions

$$\begin{aligned} f^+(s) &\stackrel{\text{def}}{=} \max\{f(s), 0\} \\ f^-(s) &\stackrel{\text{def}}{=} \max\{-f(s), 0\} \end{aligned}$$

so that  $f = f^+ - f^-$  and  $|f| = f^+ + f^-$ . The **integral** of  $f$  with respect to  $\mu$  is

$$\int_S f d\mu \stackrel{\text{def}}{=} \int_S f^+ d\mu - \int_S f^- d\mu,$$

using Definition 0.21 for each term, when at least one of the two terms is finite; otherwise, the integral of  $f$  is undefined. ( $f^+$ ,  $f^-$ , and  $|f|$  are measurable when  $f$  is (Billingsley 1995, Theorem 13.3).)

(Billingsley 1995, sec. 15)

**Example 0.27** (An integral with positive and negative parts). For the counting measure  $\mu$  on the die rolls  $D$ , let  $f(x) = x - 3$ . Its values at  $1, \dots, 6$  are  $-2, -1, 0, 1, 2, 3$ , so:

- $f^+$  has values  $0, 0, 0, 1, 2, 3$ , and  $\int_D f^+ d\mu = 6$  by Example 0.26;
- $f^-$  has values  $2, 1, 0, 0, 0, 0$ , and  $\int_D f^- d\mu = 3$ ;
- $\int_D f d\mu = 6 - 3 = 3$ , which is also  $\sum_{x \in D} (x - 3) = 21 - 18 = 3$ .

**Definition 0.23** (Absolutely integrable function (integrable function, absolute integrability)). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#). A [measurable function](#)  $f : S \rightarrow \mathbb{R}$  is **absolutely integrable** with respect to  $\mu$  (often just called **integrable**) if the integral of its [absolute value](#) is finite:

$$\int_S |f| d\mu < \infty$$

The property itself is called **absolute integrability**.

*Remark 0.11* (An absolutely integrable function has a finite integral). If  $f$  is absolutely integrable, then  $\int_S f^+ d\mu$  and  $\int_S f^- d\mu$  are both finite, because  $f^+ \leq |f|$  and  $f^- \leq |f|$ , so each sum in Definition 0.21 for  $f^+$  or  $f^-$  is at most the matching sum for  $|f|$ . So  $\int_S f d\mu$  (Definition 0.22) is a real number.

**Example 0.28** (Absolutely integrable and not).

- For the counting measure on the die rolls  $D$ ,  $f(x) = x - 3$  (Example 0.27) is absolutely integrable:  $\int_D |f| d\mu = 2 + 1 + 0 + 1 + 2 + 3 = 9 < \infty$  by Example 0.26.
- For the counting measure on  $\mathbb{N}$ , the constant function  $h(n) = 1$  is not absolutely integrable, because  $\int_{\mathbb{N}} |h| d\mu = \int_{\mathbb{N}} h d\mu = \infty$  (Example 0.26).

## $L^p$ spaces

**Definition 0.24** ( $L^p$  space and  $L^p$  norm ( $L^p(\mu)$ ,  $\mathcal{L}^p$ )). Let  $(S, \mathcal{S}, \mu)$  be a [measure space](#). For any real number  $p \geq 1$ , the  $L^p$  **norm** of a [measurable function](#)  $f : S \rightarrow \mathbb{R}$  is:

$$\|f\|_p \stackrel{\text{def}}{=} \left( \int_S |f|^p d\mu \right)^{1/p}.$$

The space  $\mathcal{L}^p(S, \mathcal{S}, \mu)$  consists of all measurable functions  $f : S \rightarrow \mathbb{R}$  for which  $\|f\|_p < \infty$ . To satisfy the definiteness axiom of a [norm](#) ( $\|f\|_p = 0$  if and only if  $f = 0$ ), functions that are equal [almost everywhere](#) ( $\mu$ -a.e.) are identified into equivalence classes. The resulting normed vector space is the  $L^p$  **space**, denoted  $L^p(S, \mathcal{S}, \mu)$  (or simply  $L^p(\mu)$ ).

For  $p = \infty$ , the **essential supremum norm** (or  $L^\infty$  **norm**) is:

$$\|f\|_\infty \stackrel{\text{def}}{=} \inf\{C \geq 0 : \mu(\{s \in S : |f(s)| > C\}) = 0\}.$$

The space  $L^\infty(S, \mathcal{S}, \mu)$  consists of equivalence classes of essentially bounded measurable functions.

**Example 0.29** (An  $L^1$  function not in  $L^2$ ). Consider the interval  $(0, 1)$  equipped with the Borel  $\sigma$ -algebra (Definition 0.4) and Lebesgue measure  $\mu$  (Definition 0.15). Define  $f : (0, 1) \rightarrow \mathbb{R}$  by  $f(x) = x^{-1/2}$ .

1. For  $p = 1$ :

$$\begin{aligned}\|f\|_1 &= \int_0^1 x^{-1/2} dx \\ &= [2x^{1/2}]_0^1 \\ &= 2 \\ &< \infty,\end{aligned}$$

so  $f \in L^1((0, 1), \mu)$ .

2. For  $p = 2$ :

$$\begin{aligned}|f(x)|^2 &= (x^{-1/2})^2 \\ &= \frac{1}{x}.\end{aligned}$$

The integral is:

$$\begin{aligned}\int_0^1 \frac{1}{x} dx &= \lim_{\varepsilon \rightarrow 0^+} [\ln x]_\varepsilon^1 \\ &= \lim_{\varepsilon \rightarrow 0^+} (-\ln \varepsilon) \\ &= \infty.\end{aligned}$$

Therefore,  $\|f\|_2 = \infty$ , so  $f \notin L^2((0, 1), \mu)$ .

This shows that  $L^2 \subset L^1$  on finite-measure sets: higher exponents impose stricter integrability requirements near singularities.

**Theorem 0.5** (Holder's inequality). Let  $(S, \mathcal{S}, \mu)$  be a *measure space*, and let  $p, q \in [1, \infty]$  be conjugate exponents satisfying  $\frac{1}{p} + \frac{1}{q} = 1$  (with the convention that  $q = \infty$  when  $p = 1$ , and  $q = 1$  when  $p = \infty$ ). If  $f \in L^p(\mu)$  and  $g \in L^q(\mu)$  (Definition 0.24), then their product  $fg$  is *absolutely integrable* ( $fg \in L^1(\mu)$ ) and satisfies:

$$\|fg\|_1 = \int_S |fg| d\mu \leq \|f\|_p \|g\|_q.$$

When  $p = q = 2$ , this inequality is the Cauchy-Schwarz inequality for integrals.

**Theorem 0.6** (Minkowski's inequality ( $L^p$  triangle inequality)). Let  $(S, \mathcal{S}, \mu)$  be a *measure space* and let  $1 \leq p \leq \infty$ . For any functions  $f, g \in L^p(\mu)$  (Definition 0.24), their sum  $f + g$  is in  $L^p(\mu)$  and satisfies:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Together with absolute homogeneity  $\|cf\|_p = |c| \|f\|_p$  and definiteness, Minkowski's inequality establishes that  $L^p(\mu)$  satisfies all axioms of a **normed vector space**.

**Example 0.30** (Applying Holder's inequality to finite sums and expectations).

1. **Finite vectors (Cauchy-Schwarz and taxicab bounds):** Equipping a finite set with the **counting measure**, Holder's inequality for conjugate exponents  $p, q$  gives:

$$\sum_{i=1}^n |x_i y_i| \leq \left( \sum_{i=1}^n |x_i|^p \right)^{1/p} \left( \sum_{i=1}^n |y_i|^q \right)^{1/q}.$$

Setting  $p = 1, q = \infty$  gives  $\sum_{i=1}^n |x_i y_i| \leq \|\tilde{x}\|_1 \|\tilde{y}\|_\infty$ . Setting  $p = q = 2$  gives the standard vector **Cauchy-Schwarz inequality**.

2. **Covariance bound for random variables:** On a probability space with measure  $P$  (Definition 0.19), taking  $p = q = 2$  gives  $E[XY] \leq \sqrt{E[X^2]} \sqrt{E[Y^2]}$ , guaranteeing that the covariance between any two square-integrable random variables is finite.

*Remark 0.12* ( $L^p$  spaces and norms in probability and machine learning). In probability, statistics, and machine learning,  $L^p$  spaces structure the geometry of random variables, estimators, and loss functions:

1. **Random variables, moments, and inclusion of  $L^p$  spaces:** When  $\mu = P$  is a probability measure (Definition 0.19) with total measure  $P(S) = 1$ , an  $L^p$  function is a random variable  $X$  with finite  $p$ -th moment:

$$\|X\|_p = (E[|X|^p])^{1/p}.$$

By **Jensen's inequality**, if  $1 \leq p \leq q \leq \infty$ , then  $\|X\|_p \leq \|X\|_q$ , so  $L^q(P) \subseteq L^p(P)$ . For instance, every random variable with finite variance ( $X \in L^2$ ) has finite expectation ( $X \in L^1$ ).

2. **The unique Hilbert space  $L^2$  and conditional expectation:** Among all  $L^p$  spaces, only  $L^2(\mu)$  is an **inner product space**, equipped with the inner product  $\langle f, g \rangle \stackrel{\text{def}}{=} \int_S fg d\mu$  (all other  $L^p$  spaces fail the parallelogram law). The completeness of  $L^2$  makes it a Hilbert space. In regression, the conditional expectation  $E[Y | X]$  is the orthogonal projection of  $Y \in L^2$  onto the closed subspace of  $L^2$  functions measurable with respect to  $X$ , minimizing the expected mean squared error  $\|Y - g(X)\|_2^2$ .

3. **Loss functions, estimation, and robustness:** The choice of  $p$  in  $L^p$  norms defines loss functions for optimization and regression:

- $L^2$  **loss (mean squared error)**: penalizes large errors quadratically, yielding the conditional mean; sensitive to outliers.
- $L^1$  **loss (mean absolute error)**: penalizes errors linearly, yielding the conditional median; robust to heavy tails and outliers.
- $L^\infty$  **loss (Chebyshev criterion)**: minimizes the worst-case maximum error across all instances.

## Product measures

**Definition 0.25** (Product  $\sigma$ -algebra). Let  $(S, \mathcal{S})$  and  $(T, \mathcal{T})$  be measurable spaces. A measurable rectangle is a Cartesian product  $A \times B$  with  $A \in \mathcal{S}$  and  $B \in \mathcal{T}$ . The product  $\sigma$ -algebra  $\mathcal{S} \otimes \mathcal{T}$  is the  $\sigma$ -algebra on  $S \times T$  generated by the measurable rectangles.

**Example 0.31** (Product  $\sigma$ -algebras).

- Give the die rolls  $D$  the  $\sigma$ -algebra  $\mathcal{S}$  of all its subsets. Each single pair  $\{(a, b)\} = \{a\} \times \{b\}$  in  $D \times D$  is a measurable rectangle. Every subset of  $D \times D$  is a union of finitely many single pairs, and  $\mathcal{S} \otimes \mathcal{S}$  contains finite unions of its sets (Theorem 0.1), as well as  $\emptyset$ , so  $\mathcal{S} \otimes \mathcal{S}$  is the collection of all subsets of  $D \times D$ .
- The rectangle  $[0, 2] \times [0, 3]$  is a measurable rectangle for the Borel  $\sigma$ -algebras, so it is in  $\mathcal{B} \otimes \mathcal{B}$ .

**Definition 0.26** (Product measure). Let  $(S, \mathcal{S}, \mu)$  and  $(T, \mathcal{T}, \nu)$  be measure spaces with  $\sigma$ -finite measures  $\mu$  and  $\nu$ . The product measure  $\mu \otimes \nu$  is the measure on the product  $\sigma$ -algebra  $\mathcal{S} \otimes \mathcal{T}$  that gives each measurable rectangle the product of the measures of its sides:

$$(\mu \otimes \nu)(A \times B) = \mu(A) \nu(B) \quad \text{for all } A \in \mathcal{S} \text{ and } B \in \mathcal{T},$$

with  $0 \cdot \infty = \infty \cdot 0 \stackrel{\text{def}}{=} 0$  and  $x \cdot \infty = \infty \cdot x \stackrel{\text{def}}{=} \infty$  for  $x > 0$ . Exactly one measure has these values (Billingsley 1995, Theorem 18.2).

**Example 0.32** (Product measures).

- For the counting measure  $\mu$  on the die rolls  $D$ ,  $(\mu \otimes \mu)(A \times B) = |A||B|$ , the number

of pairs in  $A \times B$ . For example, the set of pairs in which both rolls are even is  $\{2, 4, 6\} \times \{2, 4, 6\}$ , with  $(\mu \otimes \mu)(\{2, 4, 6\} \times \{2, 4, 6\}) = 3 \cdot 3 = 9$  pairs.

- For Lebesgue measure  $\lambda$  on  $\mathbb{R}$ ,  $(\lambda \otimes \lambda)([0, 2] \times [0, 3]) = 2 \cdot 3 = 6$ , the area of the rectangle.

## Further reading

- Billingsley (1995) develops measure theory and builds probability on it. Its early chapters define  $\sigma$ -algebras, measures, and their basic properties.
- Folland (1999) is a graduate text on measure and integration; its first chapters treat  $\sigma$ -algebras and measures.
- Gut (2013) is a graduate course in probability that opens with a chapter on introductory measure theory.

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